LAYR

Design-Led Exploration of Consumer Hardware Through Desktop Manufacturing

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Abstract

This project creates a brand identity for a niche retro-modern hardware startup company through a physical prototype and digital animation. The primary objective of this project is to adhere to current niche market needs and fully implement this design through digital and physical touchpoints.

The project details the creation of designs across physical and digital mediums, culmination of multimedia skills, and how they can be utilised to solve business needs. Leveraging current and more dated techniques, the project showcases that a single creative problem solver can elevate the design and appeal of products.

Declaration

I, Daniel Canning (20102568), I declare that this report and the work described within it are my own, except where otherwise stated. This work has not been submitted previously, in whole or in part, for any other academic award.

Introduction

This project explored speculative product design, grounded in real manufacturing logic, of a retro-modern brand ecosystem through the integration of digital and physical design practices. Primarily, the outcome of the project was the conceptual development of a design-driven hardware product. This was presented alongside supporting digital artefacts, collectively communicating brand identity, product intent, and user experience. Rather than focusing on full technical product engineering, the project prioritised design, visual language, and interdisciplinary workflows across varying mediums.

At the core of the project was a compact audio-related physical product, designed to act as a tangible anchor for the wider brand. This object was explored primarily through parametric CAD modelling and 3D visualisation, allowing for detailed consideration of form, proportions, physical interaction, and manufacturability without the constraints of full electronic implementation. The physical design was supported by an animated product showcase created in Blender, which communicated assembly logic, material intent, and importantly acted as a bridge between the physical world of 3D printing, and the contemporary world of branding and design, the digital plane.

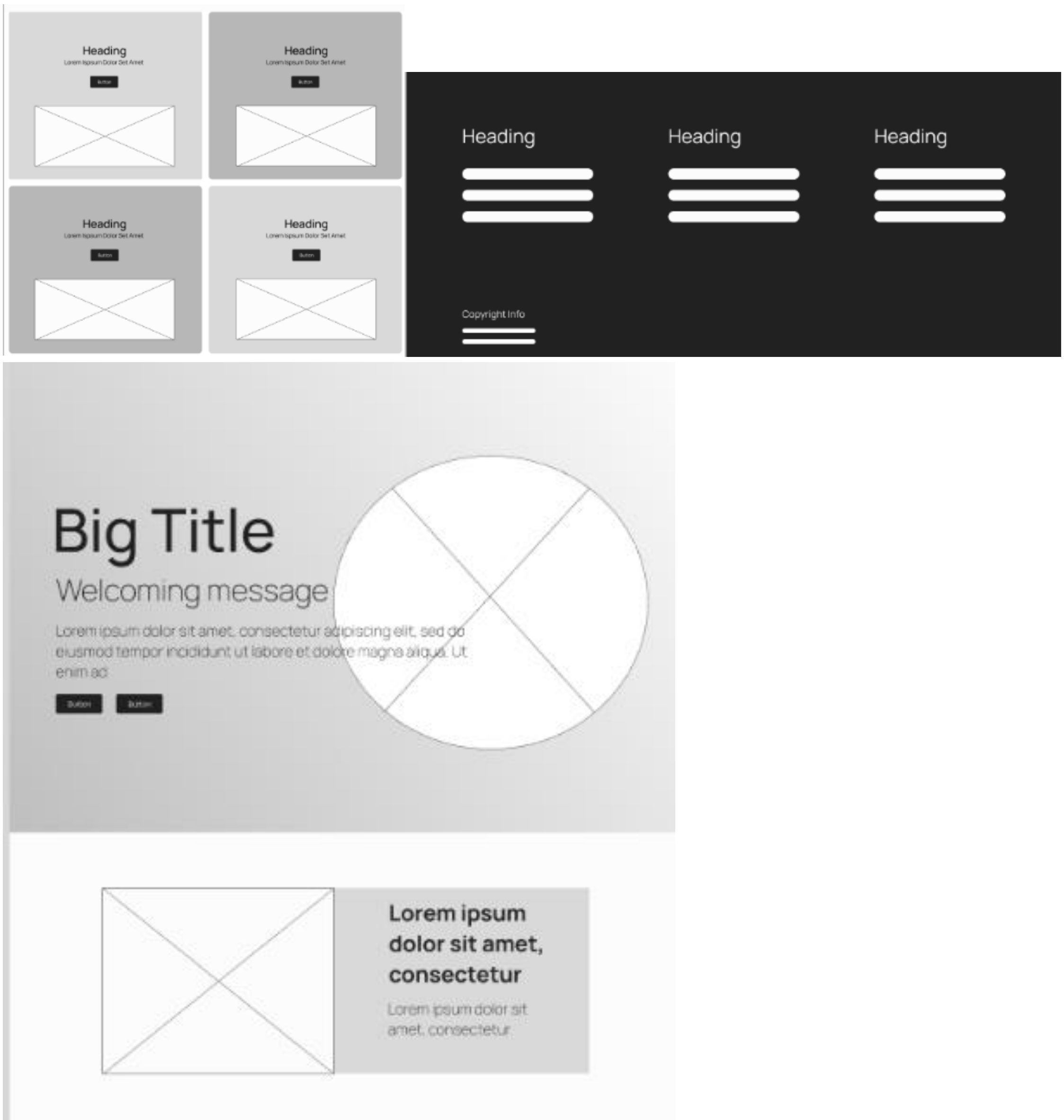
A digital platform was developed alongside the physical design outputs. This website functioned as a curated showcase rather than a fully developed application, contextualising the product design, animation, and broader brand within a single, cohesive narrative. Together, the CAD models, animation, and web showcase formed a unified demonstration of how a small, design-focused hardware brand could communicate value and identity across both physical and digital touchpoints.

Web Design, Prototyping, and Website Development

The responsive static website developed as part of this project functioned as a unifying digital layer, contextualising the physical product design, branding, and animated outputs within a single, coherent narrative space. Rather than acting as a conventional commercial website or application, it was conceived as a curated digital showcase: a place where the various strands of the project could come together, reinforce one another, and be experienced as a cohesive brand ecosystem. In other words it acts as a polished presentation layer on top of the other hard outputs.

Figma Wireframing and Layout Planning

Prior to implementation, the responsive website was planned using low-fidelity wireframes in Figma. This stage focused on structure, hierarchy, and flow. The wireframes defined key page sections such as the hero area, modular content blocks, informational cards, and supporting sections for documentation and downloadable resources.



This planning stage was essential in preventing scope creep during development. By resolving layout decisions early, the subsequent build phase could focus on visual refinement and interactive elements rather than restructuring content.

The wireframes emphasised clear visual hierarchy, generous spacing, and restrained composition. Large hero areas were used to foreground key visual material, while secondary sections adopted repeatable card-based layouts that could scale incrementally if the site were expanded in the future.

Brand Integration: Colour, Typography, and Logo Usage

The website directly incorporated the established brand identity developed earlier in the project. Colour choices were derived from the branding phase and applied consistently across backgrounds, typography, and interface elements. Vibrant gradients formed the primary canvas, the logo and animation punctuating key interactive elements without overwhelming the content.

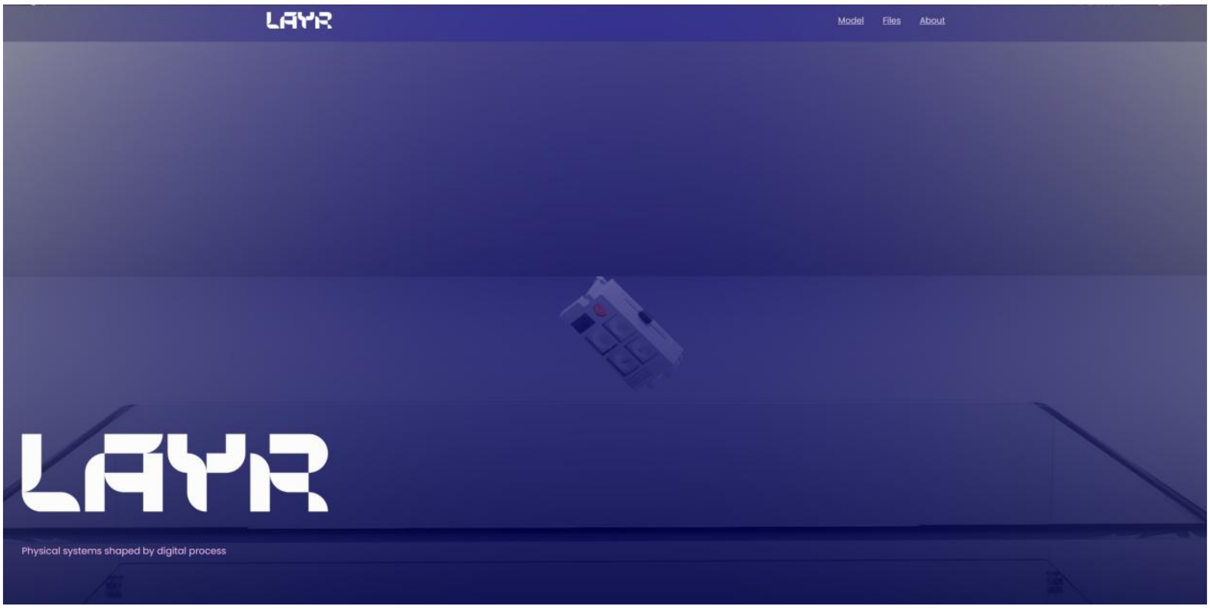
The logo was integrated as a dominant branding mark. Its placement reinforced the identity of the project without detracting from the physical artefacts and animations being showcased. This approach mirrored the project's broader design ethos, where branding exists to support clarity and intent.

Typography choices (Poppins was used) favoured readability and contemporary minimalism, aligning with the retro-modern positioning of the brand. Clear typographic hierarchy allowed dense technical explanations and lighter narrative sections to coexist comfortably on the same platform.

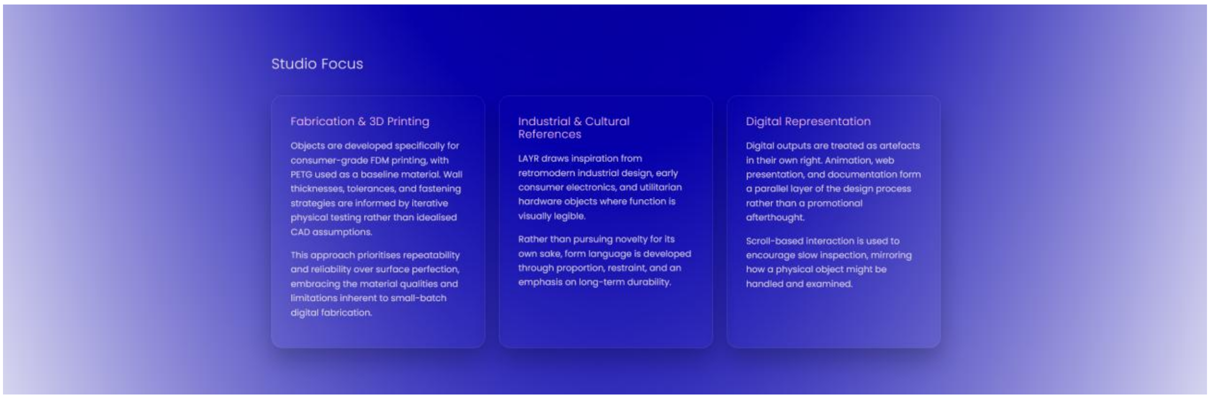
Visual and Interactive Features

Several focused blocks and visual features were implemented to strengthen the connection between the digital platform and the physical design work:

- ### Hero Section Animation
  The final product animation was embedded within the hero section, immediately anchoring the site in the core output of the project. This animation acted as an entry point into the brand and product narrative, communicating form, material intent, and assembly logic before any explanatory text was encountered.



- ### Glassmorphic UI Elements
  Interface components such as cards and content panels employed a glassmorphic design treatment. Semi-transparent surfaces, soft blurring, and subtle elevation were used to create a layered interface that visually echoed the glossy, polished aesthetic of contemporary consumer hardware, and blended well with the gradient background effects. This treatment helped soften transitions between sections while maintaining legibility and contrast.



- ## Scrubbable Scroll-Based Animation

  A rotating product animation was implemented that scrubbed in response to scroll position. Rather than playing passively, this interaction allowed the user to control the animation through natural page navigation, reinforcing the sense of tactile engagement that underpins the physical design work. This is the standout feature that directly bridges animation, interaction design, and storytelling.

- ## Downloadable Resources

  The website also functioned as a practical distribution platform. Downloadable resources, including the actual STL files generated during the project, were made available directly through the site. This reinforced the project's alignment with small-scale manufacturing, DIY culture, openness, and knowledge sharing;

transforming the digital artefact from a purely presentational surface into an extension of the making process.

Role of the Website in the Overall Project

The website's primary value lies in its ability to tie together disparate outputs into a single, legible whole. The CAD models, physical prototypes, animation, and branding were all developed using different tools and workflows; the website provides the narrative structure that allows them to be understood as parts of one system rather than isolated artefacts.

By situating the physical design within a branded, interactive digital context, the site demonstrates the way in which contemporary hardware concepts are increasingly experienced: as much through screens as through physical interaction. In this sense, the website is not supplementary but integral: it completes the project by providing the interpretive layer through which the work is framed, understood, and communicated.

While its scope was intentionally limited compared to a full production website, the digital platform successfully fulfilled its role as a contextualising artefact. It reinforces the project's design-led ethos, supports its manufacturing-aware narrative, and demonstrates how a small, independent hardware brand might realistically present and distribute its work across digital and physical domains.

Scope and Constraints

The scope of the project was deliberately constrained to reflect the academic context on the project and limited timeframe available for development and learning of new software and skills. While the full initial concept imagined a brand that aimed to produce full-scale physical hardware for consumers, the project did not aim to produce a perfect, fully functional device.

Instead, its position in the scheme of the project was as a design prototype, with emphasis on form, interaction, presentation, and conceptual brand goal adherence. More than this it served as a platform to learn manufacturing-aware design skills, especially surrounding 3d printing and its creative possibilities within a creative computing context.

With these restrictions at the forefront, the final outputs of the project were:

- Parametric CAD models of the physical product enclosure and internal layout
- A 3D animated product showcase demonstrating the designs intent
- A website set to contextualise the pieces within the guise of a brand concept

These outputs were chosen to maximise the design clarity and narrative strength of the project under time pressure, whilst still boring true to the original ideas developed in semester 1. Advanced features such as custom PCB design, full electronic integration, and large-scale manufacturing considerations were identified early as being out of scope, narratively for the project, and physically for the magnitude of learning. Therefore, they were addressed only at the speculative level.

Risk Management

The major risks associated with the project related to scope creep, technical overreach, and time limitations. Given the interdisciplinary nature of the work, decisions were carefully considered to adhere as much as possible to the vision for the project whilst simultaneously avoiding associated risks. Many decisions were required to prevent the project from becoming overly complex, technically unfocused, or adding implausible features; due to their associated added time.

Some of these risks, explicitly risks associated with the physical design elements, were mitigated by locking core deliverables early, prioritising technical and conceptual exploration over functional electronics, and favouring representational prototypes over fully realised builds.

The risk allaying approach taken was: design-first, narrative-driven. Tools such as FreeCAD and Blender were selected for their ability to support iterative refinement, in keeping with the chosen methodology (though this comes with its own risks); cross compatible formats, and high-quality visual output. The danger of the more open-ended iterative approach was again scope-creep and delays due to revisions and/or refinements. The way this was managed was paramount, as it isn't really a case of *if* rather than *when* this would inevitably happen. Accounting for this, days were added to the schedule beforehand, the iterative element remaining flexible but kept in reasonable measure.

By focusing on a set of clear deliverables rather than a wide range of half completed features, the project had clarity of intent, controlled scope management, and professional presentation within reasonable-student-project bounds.

In terms of the technical and physical risks associated with the project there were considerations to be made in terms of software compatibility, software failures, hardware compatibility, hardware failures, file storage, user error, and those in similar veins.

Potential risks were identified as translating file formats across software, hardware not being powerful enough to run programmes or create planned elements, file corruption, accidental deletion, and hardware failures. Many of these were eliminated when choosing the main software, namely Blender and FreeCAD, and using the glTF / GLB (.gltf / .glb) file format, which is cross compatible and feature rich.

Otherwise, the main files were protected using the 3-2-1 backup rule, whereby 3 copies of the data are made, one primary working copy and two secondary backup copies. Two of these are physical, one on the SSD of my machine and one on a separate physical storage device (in this case a hard drive on a separate machine), and another copy on the cloud. This seriously reduces the risk of losing files provided the backups are regularly updated. Some may see this as overkill and aim for a more manageable system of a copy on your machine and one on the cloud. However due to spotty Wi-Fi, when the odd file corruption did occur (particularly in FreeCAD) the secondary physical copy did come in handy and reduced downtime when data loss occurred

Another risk not often considered but worth mentioning is the potential health risks of melting thermoplastics in close proximity to oneself. While often debated, it is commonly understood in the 3D printing community that exposure should be limited. To combat this good ventilation and airflow were introduced to the printing space. While these did negatively affect the project in terms of added elements to battle with 3D printing, the potential risks were not worth ignoring.

Approach/Considerations

Hardware and Software Considerations

Hardware and Software were treated primarily as enablers for visualisation, interaction, and creation rather than as ends in themselves. While the product concept references real electronic components and plausible hardware architectures, the focus throughout remained on representing physical constraints, spatial relationships, and user interaction. The primary software used were FreeCAD and Blender

FreeCAD (Wiki, 2026) was used for parametric modelling of the physical enclosure and internal layout, allowing dimensions, wall thicknesses, clearances, and tolerances to be iterated quickly (at least with respect to the learning curve) and efficiently (Jorge D. Camba, 2016) as the design evolved. This parametric workflow supported rapid refinement and ensured that design decisions remained grounded in realistic physical constraints.

This was especially important when taking the 3d printing elements into account. Volumetric/Direct modelling has its place but would lead to a lot of wasted time when (ultimately) guessing dimensions, tolerances, real world fittings such as screw mounts, and manufacturing process limitations. Features created are quantifiably physically valid as the software stops you from creating impossible physical geometry. While it is possible to emulate these features in Blender, it is ultimately not the same as using a dedicated parametric modelling software. Moreover, you would be missing out on extra features such as technical drawings and FEM analysis.

Blender was however used for high-quality rendering and animation, something the software is highly tuned for; enabling the product to be presented in a manner consistent with contemporary hardware concept showcases. While it is not beginner friendly, it is highly optimised, customisable, and produces good results.

Another important note is that both these software are free and open source, something that I personally value and support from a budgetary perspective and moral one. Being able to get access to high quality software for free is a huge blessing for the project and is one of the contributing factors to its viability.

Some other supplementary software included Osci-render and Figma. Osci-render was used as part of the video animation to add a layer of visual flare and experiment with a niche technology from a creative computing angle. Osci-render acts as an oscilloscope emulator (Ball, 2026), primarily for making music as a synthesiser type tool. However here it was used to generate oscilloscope graphics through blender.

Figma played a small role in the web development side of things to create wireframe mock-ups, giving a plan before starting real development work

This software stack was selected to reflect industry-relevant workflows while remaining accessible within an academic framing. Open-source and widely adopted tools were prioritised reducing technical risk, ensuring file compatibility, and supporting the iterative, experimental approach across elements designed around the digital and physical.

Hardware list

- **Bambu labs A1 mini with 0.4mm nozzle** – *Used for 3D printing*
- **Primary Machine (Custom PC)**

Processor (CPU)	Amd Ryzen 5 7600, 6-core, 12-thread processor with a base clock speed of 3.8 GHz and a boost clock speed of up to 5.1 GHz
Motherboard	asrock b650m-hdv/m.2
RAM	32-GB (2x 16-GB) DDR5 6400 CL36 Corsair Vengeance
Storage	1TB SSD (Fanxiang)

Operating System	Windows 10, later in development Windows 11
Graphics Card	ASUS Dual OC 5070 12GB

- **Creality Space Pi Filament Dryer Box**
- **Wacom Drawing Tablet** – Used in logo design phase

Software list

- FreeCAD (V0.2.1)(V1.1) – Opensource parametric modelling software
- Blender (V4.2)(V5.1) – Opensource direct modelling software, also video editor, VFX, and many other creative elements included
- Adobe Illustrator 2025 – Vector design Software
- GitHub – Used for hosting
- Osci-render – Visual Flare
- Premier Pro – Video Editing
- Visual Studio Code – Web development platform

Similar Projects and Design Precedents

The project was informed by a range of historical and contemporary design precedents spanning consumer electronics, hardware brands, and small-scale DIY electronics projects. Inspiration was taken from the work of Teenage Engineering, whose products demonstrate restrained, simplified interfaces; strong industrial design, and cohesive visual language, elevating simple electronic functionality into culturally significant objects (Kostiuk, 2024). Their emphasis on playful minimalism, legibility of interaction, and material honesty strongly influenced the approach taken to form, control layout, and presentation.

Historical references were also drawn from the Nokia Design Archive, which provided insight into an earlier era of consumer electronics where physical interaction, robustness, and clarity of form were central to product identity. These artefacts highlighted the value of tactile controls, purposeful geometry, and long-term usability (Barandy, 2024). Principles that remain relevant when reinterpreted through a contemporary, retro-modern lens. Their internal documents and workshop manuscripts proved to be an invaluable source of information surrounding the project and was a great inspiration when it can to approaches taken to design across the spectrum (Aalto University, 2026).

In addition to commercial precedents, the project also examined a wide range of small-scale DIY electronics projects and kits, particularly those distributed within hobbyist and maker communities. These projects were valuable in understanding the potential consumer group, as well as how similar, smaller scale versions of my own physical elements of the project were handled by those in the community (Formlabs, n.d.). However, this research process also revealed significant challenges. The 3D printing and maker ecosystem is heavily saturated with low-quality reference material and poorly documented projects, many of which prioritise novelty production over considered design choices. Often these creators do not even attempt to optimise their output for the medium of 3D printing, highlighting what would be large design consideration in my own project. More recently, this issue has been compounded by the increasing prevalence of AI-generated artefacts, which often present visually convincing but physically lacklustre designs in terms of 3D printing output specifically.

As a result, it proved difficult to identify reference projects at a similar scale that balanced aesthetic quality, physical plausibility, and clear documentation. This scarcity reinforced the importance of grounding the project in established design principles rather than relying solely on contemporary maker, or "YouTuber" trends. It also influenced the decision to prioritise manufacturing-aware design, abstraction, and parametric control, ensuring that the resulting product concept remained credible despite operating within a highly speculative and visually noisy design space.

Component Selection and Abstraction

Although the project did not aim to implement a fully functional electronic system, real-world components had to be considered during the design process to ensure plausibility and appropriate scale. Components such as a microcontroller, display, battery, and physical controls informed internal volume requirements and interaction design, even when represented in a slightly abstracted form.

The decision to abstract electronic components was intentional. By avoiding technical detail in elements outside the scope, the project could keep focus on enclosure design, interaction, and presentation while still demonstrating an understanding of how a design is influenced by real world elements for such a product to be realised in practice (Crosley, n.d.). This abstraction also supported the conceptual distinction between the DIY kit and a hypothetical premium version utilising a custom PCB, which could consolidate functionality into a slimmer and more refined internal architecture.

Cost and Design Constraints

Cost and resource constraints played a significant role in shaping the strategy adopted for physical prototyping during the project. As the physical artefact was developed primarily as a learning platform for manufacturing-aware design, it was necessary to be selective and strategic about physical prototyping to minimise both material costs and general wear on the 3D printing equipment.

Rather than relying on repeated full-scale prints, the design process prioritised incremental testing of smaller elements. For example, instead of printing a full iteration of a design to test custom screws, a simple, short, cylinder was printed with matching thread to the large model iteration, and the large model was subsequently changed based on the findings. Digital iteration, and targeted physical prototypes, focused on specific problem areas such as wall thickness, fit tolerances, and enclosure splits to narrow focus on problem areas and simplify the design process. This did have the unfortunate side effect of increasing modelling times by adding extra test models, however saved more in print times.

Material choice was also heavily influenced by the practical limitations of the available hardware (Bambu Lab A1 Mini) and the relative cost of consumables. While ABS would've been a strong candidate due to its durability, heat resistance, suitability for functional enclosures, and very effective post processing options, its higher material cost, increased failure rate, stricter environmental requirements, and the necessity for a different printer, made it impossible within the constraints of the project.

The two other thermoplastics that were evaluated for the enclosure: PLA (Polylactic Acid) and PETG (Polyethylene Terephthalate Glycol) both of which are commonly available and can be printed by a wide range of FDM devices.

- PLA offers excellent printability, dimensional accuracy, low warpage, and cheap cost (Mohammed Algarni, 2021). These advantages make it a strong candidate for our use case. However, PLA's drawbacks for a handheld, regularly used device include lower impact resistance, increased brittleness over time, especially with exposure to UV rays, and susceptibility to deformation at moderately elevated temperatures (e.g., a warm vehicle interior or prolonged direct sun exposure). PLA can also exhibit creep under sustained stress in thin fastening features.

- PETG provides enhanced toughness and impact resistance, improved fatigue behaviour for parts subject to repeated handling, and a higher glass transition temperature, which better resists softening in warm conditions (Mehmet Albaşkara, 2025). These properties are advantageous for screw bosses, snap-fit regions, and thin sections subjected to assembly loads. The trade-offs are greater stringing propensity, slightly reduced dimensional sharpness, and more involved tuning to achieve optimal surface quality.

These material constraints directly informed design decisions, including wall thickness, part geometry, and clearance allowances. By treating the cost and printer limitations as active design parameters rather than obstacles, the project could show how meaningful physical products can be developed under constrained conditions without sacrificing visual or conceptual quality.

Cost constraints also limited software selection, predominantly forcing the use of free and open-source options however the impact was more greatly felt with the 3D printing outputs.

Design for 3D Printing

A key consideration throughout the physical design process was optimisation for additive manufacturing (Ala'aldin Alafaghani, 2017), specifically consumer-grade FDM 3D printing surrounding the Bambu architecture, Bambu being one of the biggest players in the consumer and smaller-scale commercial space as the time of this project. The enclosures aesthetic design was developed with attention to 3D printing constraints and optimisations, including minimum wall thickness, overhang angles, part orientation, tolerances, material selection, layer heights, ambient temperatures, heat-bed and nozzle temperatures, filleting and chamfering on the correct axes, optimising print times through the design and in the slicer, .stl export quality, and various other 3D printing specific qualities.

1 STL export quality test (reflections of light show quality from lowest to highest left to right)

Rather than treating 3D printing as a purely technical constraint, it was approached as a creative medium. Design decisions such as surface geometry, internal ribs, strength subject to layer orientation, and enclosure mating were informed by both functional requirements and aesthetic opportunities afforded by layer-based fabrication. This allowed the physical artefact to serve as a practical exploration of manufacturing-aware design, reinforcing skills that are directly relevant to creative computing and digital fabrication practices.

The emphasis on 3D printing also aligned with the broader brand narrative, which positions small-scale, accessible manufacturing as part of its identity. By considering the limits of consumer-level fabrication, the project is opened to demonstrate how chosen mediums *must* affect design outputs to create a cohesive identity.

Printer Technology

An FDM printer was used due to accessibility, material availability, and the ability to prototype functional enclosures rapidly. The workflow leveraged a standard 0.4 mm nozzle. Print temperatures, retraction, and cooling were tuned to balance PETG's stringing tendency with strong layer adhesion and acceptable surface quality.

Surface Quality Optimisation and Experimental Treatments

To improve the perceived quality and reduce visible layer lines, three techniques were evaluated on test coupons and enclosure prototypes.

Ironing (Top-Surface Refinement)

Ironing was applied on top layers to reduce voids and produce a smoother finish. By traversing the nozzle with minimal extrusion on the final layer, heat and gentle pressure promote micro-level reflow. On flat cosmetic panels, ironing yielded a notable reduction in apparent layer texture and improved tactile smoothness. Minor trade-offs included a slight increase in print time and the need to avoid overheating on small, thermally constrained features (Prusa, 2025).

Fuzzy Skin (Textured Surface for Grip)

The fuzzy skin feature was tested to introduce a fine, randomised surface texture that enhances grip and visually masks minor artefacts. While effective in producing a non-slip, tactile finish, it reduced edge sharpness around precision openings (e.g., buttons, USB port) and complicated post-processing. As well as this it drastically alters the visual styling of the final output and increases overall print time. Given the priority on dimensional precision at interfaces, fuzzy skin was not used on the final enclosure.

Annealing in a Salt Bath (Layer-Line Reduction and Stability)

Selected PETG prints were annealed in a controlled salt bath to reduce the visibility of layer lines and improve thermal stability (Supaphorn Thumsorn, 2022). The salt medium supports the part uniformly during heating, limiting warpage while permitting polymer chain relaxation and reorganisation. Post-anneal, prototypes exhibited smoother surface continuity and slightly increased rigidity. Minor dimensional changes were observed and subsequently compensated in CAD for critical features. This method demonstrated promise for improving both aesthetics and service temperature tolerance, but it requires careful control when tight tolerances are essential.

Fastening Choice: Screws vs. Click-Fit

Two fastening approaches were considered for the enclosure: a screwed-together design and a click-fit design created through tolerance control in slicer settings.

Screwed Assembly

A screwed enclosure offers consistent, reliable assembly across different printers and materials. Because it does not depend on precise user slicer calibration, it ensures that the enclosure fits securely regardless of machine variation. It also supports repeated opening for repairs or upgrades without risking wear on the joint.

Click-Fit Assembly

A click-fit design provides a clean, hardware-free experience, relying on tolerances influenced by slicer settings such as horizontal expansion and extrusion compensation. While this can create an elegant snap-together interaction, performance varies significantly between users depending on printer accuracy, filament choice, and calibration quality.

Maintenance

Successes in Project Development

The project was successful in meeting its primary objectives within the defined scope. The strongest outcomes were achieved in the areas of physical product design, manufacturing-aware CAD development, and visual communication through 3D animation. The decision to prioritise a small number of well-executed deliverables resulted in a credible representation of the concept.

The parametric modelling workflow proved particularly effective, enabling very rapid iteration. It supported informed decision-making throughout the design process and aligned well with the project's emphasis on 3D printing and small-scale fabrication. The integration of Blender for rendering and animation further strengthened the final presentation and acted as a good segue from the rigid parametric workflow.

What Worked and What Did Not

The interdisciplinary nature of the project was one of its key strengths. Combining CAD, fabrication considerations, animation, and web presentation allowed the project to function as a holistic design system rather than a collection of isolated outputs. The workflow between FreeCAD and Blender was particularly effective, supporting both technical accuracy and visual storytelling.

However, not all planned aspects were realised to the same extent. The scope of the web component was reduced significantly from early ambitions, functioning primarily as a showcase rather than a fully interactive platform due to the unprecedented learning curve that was both primary software, especially the parametric side of things. Similarly, the electronic system remained abstracted, with no fully functional prototype produced.

While these elements were initially explored conceptually, time constraints and the prioritisation of manufacturing-aware design led to a deliberate narrowing of the focus.

Failures, Limitations, and Contingencies

The most significant limitations encountered related to time, resource availability, and the complexity of balancing multiple disciplines within a single project. Attempts to pursue deeper electronic implementation or broader branding and digital interaction were identified as high-risk in terms of scope creep and were subsequently cut. This resulted in some early exploratory work not being carried through to final outputs.

To mitigate these limitations, the project adopted several contingency strategies. Physical prototyping was approached selectively to reduce material waste and printer wear, relying heavily on digital validation and small-scale tests before committing to prints. The modularity of the design of the project allowed me to identify features or ideas to be cut for the sake of brevity. Design abstraction was used intentionally to maintain plausibility without requiring full technical implementation, allowing the project to progress without becoming stalled by what would be unresolved engineering challenges.

Provisions to Address Reduced Development Scope

Where development depth was limited, the project compensated through clarity of intent and strength of presentation. The animated product showcase was used to communicate interaction, assembly logic, and material qualities that would otherwise have required a fully built prototype. Similarly, the emphasis on parametric design and manufacturing constraints ensured that the physical artefact remained credible and grounded despite its representational nature.

By reframing constraints as design parameters rather than shortcomings, I believe the project managed to maintain its coherence and relevance to its original objectives. The resulting body of work demonstrates a considered balance between my unending scope ambition and realistic feasibility, prioritising depth of execution, learning opportunities, and reflective practice.

Project Analysis and Specification

Project Size

This project was undertaken as an individual final-year project, with all research, design, modelling, prototyping, animation, and documentation completed independently. The scope and scale of the work were therefore planned to be achievable by a single developer operating across multiple new and distinct disciplines, including branding, CAD, 3D printing, animation, and web presentation. The modular structure of the project allowed parallel development of different components, which was essential in managing workload and accommodating external academic and personal commitments.

Economic Feasibility

The project demonstrated a strong level of economic feasibility at a conceptual and small-scale production level. Rather than targeting mass-market manufacture, the product and brand concept were positioned within a niche, design-led hardware space, where value is derived from aesthetics and narrative rather than technical specification or economies of scale. This positioning aligns with contemporary markets for enthusiast hardware, DIY kits, and limited-run design objects.

The physical product concept was well suited to low-volume production methods such as consumer-grade 3D printing and the use of off-the-shelf electronic components. These approaches significantly reduce upfront costs, remove the need for specialist tooling, and allow production to scale incrementally based on demand. As a result, the project could plausibly be marketed as a downloadable DIY kit, a made-to-order product, or a limited batch release without substantial financial risk. This said however, it would need more development in terms of technical electronic feasibility.

In addition to direct product sales, the project outcomes were economically reusable across multiple contexts. The CAD models, fabrication workflows, branding assets, and animation pipeline developed during the project could be adapted for future products within the same brand ecosystem or repurposed for design consultancy, prototyping services, or educational use. This reusability increases the long-term value of the work beyond a single artefact.

While full commercialisation was always outside the scope of the project, the design decisions, production assumptions, and presentation strategy indicate that the concept is economically viable within a small-batch or design-conscious market.

Organisational Feasibility

The organisational feasibility of the project was considered moderate and intentionally limited by design. The product concept was developed as a largely self-contained artefact within a broader brand ecosystem, rather than as a component designed to integrate with complex external systems or existing organisational infrastructures. This approach aligned with the project's scope and with the realities of small-scale, design-led hardware initiatives.

In terms of updating and maintenance, the project did not propose a formal update pipeline. Instead, changes and improvements would occur through physical revisions, iteration of design files, or new product variants. This model is feasible within a small team or independent creator context but would not scale efficiently within larger organisational structures that require continuous deployment, remote updates, or user data management.

The product concept was not designed to integrate directly with existing digital platforms or enterprise systems. Rather, it functioned independently, supported by a lightweight web presence intended for presentation and documentation rather than operational dependency. While this limits compatibility with larger ecosystems, it reduces organisational complexity and aligns with the project's emphasis on physical artefacts and manufacturing-aware design.

The project does have organisational feasibility within the context of small-scale production, independent creators, or niche design-led brands. However, it was not intended to function within, or adapt to, large-scale organisational systems without significant restructuring. This limitation was accepted as an appropriate trade-off given the project's goals, scope, and academic context.

Technical Feasibility

The technical feasibility of the project presented a significant challenge due to my limited prior experience with essential all of the core tools and processes required. Aside from some familiarity with Autodesk Maya, which provided a partial conceptual foundation for working in Blender, most of the technical workflows, especially parametric CAD modelling, consumer-grade 3D printing, fabrication-aware design, and product-focused animation; were entirely new at the outset of the project. This steep learning curve represented the largest technical hurdle encountered during development and was the cause of most of the delays in the pipeline.

FreeCAD was used for parametric enclosure design despite no previous experience with parametric or constraint-based modelling software. Developing competency in sketch constraints, feature dependencies, and dimension-driven design required extensive experimentation and reference to documentation. Similarly, Blender introduced new technical demands related to lighting, materials, camera control, and animation that differed substantially from prior experience in other 3D modelling tools. These challenges were compounded by the integration of Osci-render within the video production pipeline, which required additional learning to achieve the desired visual style and rendering outcomes.

Physical prototyping through FDM 3D printing added a unique layer of technical complexity. From the foundation of the design the project required hands-on understanding of printer behaviour, material properties, slicing parameters, and tolerance variation. All of these being areas in which I had no prior practical experience. Iteration was therefore slower initially, and early design decisions often required revision as fabrication constraints became better understood and errors were ironed out. This contributed directly to the decision to scale back and refine the scope originally defined in the Semester One report, prioritising my execution of elements over the original bloated list of features.

Supporting tools such as Figma and Adobe Illustrator were used in a limited capacity to create pieces to conceptualise the project as a whole, namely layouts, branding elements, and presentation assets. While I had prior experience with these tools, their role in the project was intentionally minimal and supportive, serving primarily to facilitate communication and visual planning for the CAD and video outputs rather than acting as core development platforms. They also helped ease me into the more difficult segments of the project where I was stepping into the unknown so to speak.

Despite these challenges, the project ultimately demonstrated technical feasibility through my sustained learning and adaptation. Over the course of development, what had been unfamiliar tools became instruments for design iteration, fabrication-aware modelling, and visual storytelling. Yes, the learning curve necessitated scope refinement, but it also resulted in a substantial expansion of my technical capability across even mediums I had been familiar with by providing a new lens to view design through. In hindsight, the technical difficulty of the project was significant but highly rewarding, contributing meaningfully to both the final outcomes and my broader skill development exponentially, within a creative computing context.

Methodology

The Problem from a Process Viewpoint

At inception, the project involved the development of a brand identity spanning digital and physical mediums. From a process perspective, the primary challenge lay in balancing an open-ended creative workflow with the need for sufficient structure to reliably produce tangible deliverables and stary on track. The interdisciplinary nature of the work required constant negotiation between exploration and execution, particularly when moving between conceptual design, physical prototyping, and digital presentation.

Although high-level milestone deliverables were defined early in the project, the specific requirements of each evolved continuously as the work progressed. Design elements, interaction decisions, and physical constraints required iterative refinement as opposed to a more linear completion. The project demanded a process that could accommodate adaptive requirements without compromising overall direction or coherence when it came to the bigger picture.

The web component ultimately functioned as a focused showcase rather than a fully interactive system, while the physical product design was developed as a modular artefact that could be explored independently of other deliverables. This modularity proved essential in managing scope, allowing individual components, such as branding, enclosure design, animation, and web presentation; to progress in parallel. The project therefore required a methodology capable of supporting non-linear design exploration while still providing enough structure to maintain momentum and ensure completion within the available timeframe.

Assessment of Chosen Methodologies

Design Thinking was adopted as the primary methodology due to its strong alignment with the creative and iterative demands of the project. Its emphasis on empathy, ideation, and prototyping supported the exploration of multiple design directions before committing to final outputs, while maintaining a consistent focus on user experience and brand coherence (Hasso Plattner, 2016). This approach proved particularly valuable in navigating the ambiguity inherent in early concept development and in responding to constraints encountered later in the process.

The project followed the five stages of Design Thinking as outlined by the Hasso Plattner Institute of Design: Empathise, Define, Ideate, Prototype, and Test. While these stages were most visible during the initial research and concept phases, they also operated at a micro level throughout the project. Each design task cycled through these stages to ensure alignment with the overarching vision.

For example, during logo development, user expectations and aesthetic preferences were first considered to establish an empathetic understanding of the target audience. These insights were then synthesised into a clear problem definition, guiding ideation and refinement. Similar cycles occurred during physical product design, where prototyping and testing informed iterative changes to form, layout, and manufacturability.

To address the need for structure and time management, Agile principles were incorporated alongside Design Thinking. Development was organised into short, focused sprints, each targeting a specific deliverable such as CAD refinement, animation development, or web layout. This hybrid approach allowed creative exploration to continue without losing visibility of progress or deadlines. Agile practices proved particularly effective in responding to emerging constraints, such as time limitations and fabrication challenges, by enabling prioritisation and scope adjustment without derailing the overall project.

The final workflow progressed from research and concept development into iterative design and prototyping, followed by continuous testing and refinement. Rather than enforcing a rigid sequence, the methodology allowed phases to overlap and loop as needed. In theory this flexibility would've enabled the project to adapt to unforeseen challenges while still consolidating all outputs into a cohesive brand ecosystem that demonstrated synergy between digital and physical design.

Post project Analysis of chosen methodologies

Ultimately, this method wasn't viable from the perspective of solo development in the context of this semester. At least, not for this project. While design thinking is enjoyable and certainly usable, I found that its process didn't lend itself in capacity of my usage of it to the tighter timelines final semester of fourth year demanded. It was too loose to pick the project back up from whence I left it due to any outside circumstances, an inevitability of life when the project is not a sole focus for the semester.

As for Agile, the sprints worked well, however once a backlog built up and stress took over, I found it suboptimal. This would suggest that it needed more focus than was feasible to give in the planning and coordination of sprints. Agile works over structured, consistent, time-blocks, something that I do not believe most students have the luxury of conforming to. More than this solo development removes a lot of Agiles core principles, and the creative nature of the work leads to the methodologies fighting each other, rather than complimenting each other.

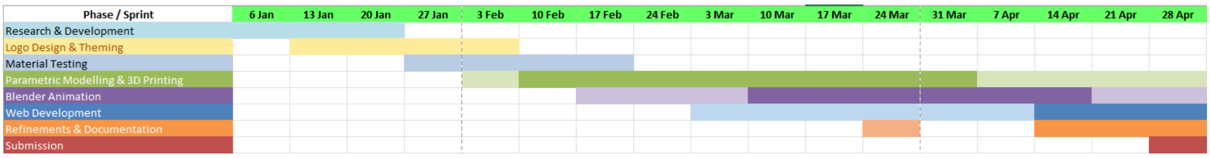
All this being said, I do believe that this hybrid approach would be fantastic if the project was a sole focus. When it did work, particularly in the earlier development stages, the two methodologies balanced each other well and the system felt as though it were aiding development as I had intended, and I do believe I will use it in the future to some effect. Ironically the modularity intended to aid development instead became overwhelming.

In retrospect, the project would have benefited from a more strongly staged process structure at a macro level at the onset. While Agile principles supported flexibility and responsiveness, the lack of rigid phase boundaries increased cognitive overhead when work was frequently interrupted.

A more sequential, stage-based approach like a Waterfall structure would likely have reduced this overhead by providing clearer "what comes next" guidance at points of re-entry. However, a strictly linear methodology would have been incompatible with the exploratory and iterative nature of design work. The experience highlighted the value of combining strong high-level structure with localised flexibility, particularly for solo creative projects operating under fluctuating availability.

Management of the Process

Gantt Chart



To keep track of deadlines and planned sprint timeboxes, a Gantt Chart was used. This method came from a project management module in the previous semester, and I find it helpful at a higher level to get an overview of time spent, and more importantly time remaining. The deeper colours represented times of heavy development, while lighter coloured overlaps occur where development was less intense and was more casual in any free time available between other engagements. The latter half of the semester was heavy on output development, and it was important to use a visual guide to keep on track and not lets sprints run wild.

Process Outline and Details

The project followed a staged development process informed by the Semester One Gantt chart, with phases functioning as loosely defined sprints rather than rigid, time-boxed iterations. Each phase focused on a major deliverable, while allowing overlap and revision where required

Phase1: Research & Concept Development

This phase spanned the first two weeks of the semester. It's areas of focus were on the conceptual brand direction, potential directions for deliverable end results, research into established brands and design research based mostly around the Nokia design archive, and supplementary tooling that could be useful such as software plugins or website.

During this phase, my main takeaways were, the setup of formal project management tools that I will discuss next, identifying the art direction of the project, discovering Osci-render and the blender plugin Blendkit, as well as a host of reading material from past documentation on the Nokia design archive. All of this helped form a better outline of the practical scope, editing the scope from first semester to address the findings.

Phase 2: Logo design and theming

After the success of the last sprint, here I was feeling good about my choices in terms of development methodology. I set about creating a host of potential logo designs and theme fitting colours and materials I would later use in the web development and Blender animation elements of the project. The logos were created using Adobe illustrator and inspiration came from a few sources. Identified in the first phase, I created some of the logos based on old logo design manuals from the 70's. However, after taking the model approach in a more modern way, I found that these didn't fit the new vision. As such, I then proceeded in a more modern direction, minimalist logotypes and the like. Once they were made it was also important to respect future phase elements when it came to colours and exporting. Knowing that I would have to use them digitally naturally all designs had to be catered for this and filetypes exported based on their purposes. This phase spanned 2 weeks and was also a success in terms of methodology and processes.

Phase 3: Material Testing, Parametric Modelling, and 3D printing

This was the first heavy development phase and where the project really started taking place. It was also where I found my methodology lacking, as discussed previously. The challenge parametric modelling presented was far greater than anticipated, and other modules had started to require attention in respect to assignments during this period. This phase could've been broken down into two phases; however, it was more efficient time wise to link the three elements together as they ultimately rely on one another. As seen in the Gantt Chart, there was some material testing done before the phases were intentionally merged

Testing the materials came first as it was apparent from the research carried out in phase 1 that my designs should be based around material selection. While initially the tests were carried out purely on the materials, once development of the parametric models started it evolved into more functional tests based around the design. Once the decision to use PETG had been solidified, material testing changed to material finish testing and refinement which was constantly refined throughout the CAD process.

CAD development presented the most significant bottleneck during this phase. Due to the nature of the work, both learning and progress were slow and were forced continue long past the original plan (see Gantt chart.) The learning of the software happened through video tutorials, official documentation, forum posts, and much trial and error as progress was continuously made and lost through various errors in the design that were not apparent purely from research.

3D printing was conducted in parallel with CAD development and testing, providing regular opportunities to validate digital decisions through physical output. It involved a lot of slicer tweaking, not to be simply functional, but to be as efficient as possible in regard to both time and material usage. Many of the optimisations in this phase were not necessary but did elevate my understanding of the process and interactions between the print materials behaviour and the hardware.

The core development of this phase lasted 8 weeks, split between the three elements contained within.

Phase 4: Blender

The Blender portion of the assignment could not have come sooner. It was a huge change from the previous phase. While the learning was still immense, taking up the bulk of the time as opposed to the physical output of the phase, the quality of the reference material and information was a proportionally gigantic upgrade over FreeCAD resources. This was easily the most pleasant and creative phase and had a lot more motivation behind it as a result. The phase also entailed some video editing and working with Osci-render, which was an interesting spin on development.

The major period of development in this phase lasted 6 weeks.

Phase 5: Web development

By the time of web development, it was a welcome change of pace, with familiar technologies involved it was a good idea to round out the development with something that required much less new information. That said it did suffer the effects of burnout and stress by this point of the semester, so it was balanced overall. This phase lasted 3 weeks.

Phase 6: Refinement and Documentation

Refinement was ongoing naturally as was the nature of the design thinking elements of the methodology however this phase was always planned as finishing touches at the end of the semester. This assumption proved optimistic in practice and there was of course still work to be done on physical outputs. The work was closer to tying up loose ends that other modules had taken priority over, rather than full on development.

Documentation was heavy going as the course generally did not have a lot of heavy written segments in any modules. It required considerable time investments and would've been better served by being consistently updated as the project progressed, instead of in its own phase.

Time allocation for phases varied depending on technical difficulty and competing academic demands; however, the staged structure provided sufficient macro-level guidance to support project progression while maintaining flexibility.

Planning Approach

Formal project management tools such as Trello boards or continuous Git-based version control were not sustained throughout the project, although an Excel Gantt chart was used for tracking sprints. Instead, planning was conducted in a lightweight and adaptive manner, shaped by the realities of solo development, frequent external interruptions, and a steep technical learning curve. This approach prioritised re-entry, clarity, and decision-making over adherence to rigid process artefacts that were not working for me.

Initial planning was established through the Semester One Analysis and Design documentation, which defined the project's original scope, ambitions, and high-level milestones. As development progressed, this plan was actively revised in response to technical difficulty, time constraints, and emerging understanding of the tools involved.

I felt, as backlogs filled up and sprints didn't line up with the outlined plans, that it was more effective to follow a dynamically evolved work structure. The backlogs didn't serve as checklists but instead a creep of perceived failures and shortcomings that realistically were not representative of the truth, or work I had been putting in. The pivot took a large weight off my mind and helped me focus more on the work at hand. This is not to imply the methodology or planning was abandoned, just that strict adherence to software-based management systems were an active hindrance during the production of this project and it was an important step during the process to recognise this and restructure.

Incremental Planning and Revision Over Time

Rather than maintaining a fixed backlog or sprint schedule, work was organised into loosely staged phases such as concept development, CAD modelling, fabrication exploration, animation, and presentation. Progression between these stages was conditional on achieving functional or conceptual stability rather than on predefined dates. This staged approach reduced cognitive overhead when returning to the project after periods of absence and helped maintain momentum despite interruptions.

Adaptation of Methodology in Practice

Agile concepts such as task decomposition and prioritisation influenced early planning; however, strict sprint structures proved unsuitable given inconsistent availability and the exploratory nature of the work. Rather than heavily enforcing sprint backlogs, planning occurred at the level of immediate next actions based on the backlog, with the priorities reassessed regularly based on technical feasibility and remaining time.

This adaptive approach allowed the project to respond effectively to unforeseen challenges, particularly in relation to learning new software, resolving fabrication constraints, and managing the integration of animation and rendering pipelines. This ensured that deliverables were completed.

Process Outcome

Although planning artefacts were informal, the project demonstrates clear evidence of process management through its coherent progression, scope refinement, and controlled convergence on final outputs. The evolution of the project over the semester reflects conscious planning decisions informed by technical reality rather than ad-hoc development. In this context, the absence of heavyweight planning tools did not indicate a lack of structure, but rather an intentional shift toward a process model better suited to solo, design-led work under fluctuating constraints.

Project Evaluation

Re-occurring Problems and Challenges

A number of re-occurring problems emerged throughout the project, most of which were rooted in the aforementioned learning curve and the realities of solo development. One of the most persistent challenges was onboarding with unfamiliar software, particularly FreeCAD. Early in development, limited experience with parametric modelling led to geometry corruption, unstable sketches, and surprisingly frequent file crashes, especially when creating technical drawings. These issues frequently required rollback to earlier versions or complete reconstruction of features, slowing progress and reinforcing the importance of disciplined modelling practices.

Physical fabrication introduced additional recurring challenges. Screw fitment and fastening strategies required multiple revisions as real-world tolerances differed from initial digital assumptions. In fact, I ended up creating several iterations of custom screws and threaded inserts. Iterative adjustment of thread clearances and boss geometry was necessary to achieve reliable assembly with FDM prints. Print bed adhesion also proved problematic at various stages, influenced by material choice, ambient temperature, bed cleanliness, and first-layer tuning. These failures were not isolated events but ongoing considerations that shaped both the design and prototyping strategy.

On the visualisation side, Blender introduced performance constraints, particularly during rendering and animation. Long render times limited iteration speed and required conscious trade-offs between visual fidelity and practical turnaround. In tandem, frequent interruptions caused by other academic responsibilities made project re-entry a recurring difficulty, often requiring time to re-establish context across CAD, fabrication, and animation workstreams.

Another notable challenge was a tendency toward over-researching tools, workflows, and reference material before committing to physical outputs. While this research supported understanding and reduced certain errors, it occasionally delayed hands-on experimentation, which ultimately proved to be the most effective learning mechanism once physical development began.

Overcoming Problems Through Adaptation

These challenges were addressed through a combination of process refinement and practical adjustment. Issues with FreeCAD stability and geometry corruption were mitigated by simplifying sketches, reducing dependency chains, and working in smaller, more modular features. Frequent versioning and conservative modelling practices reduced the impact of crashes and corruption over time.

Fabrication issues were overcome through targeted physical testing rather than repeated full-scale prints. Screw and tolerance problems were isolated using small test geometries, allowing adjustments to be validated quickly and economically. Print adhesion issues were addressed through iterative tuning of first-layer settings, surface preparation, and material-specific adjustments rather than wholesale redesign.

Performance limitations in Blender were managed by staging renders, reducing scene complexity during iteration, and reserving high-quality renders for final output. Crucially, once physical prototyping became more central to the workflow, many earlier learning difficulties resolved more efficiently. Hands-on making clarified abstract research, reduced uncertainty, and accelerated understanding in ways that documentation alone could not.

Immediate and Long-Term Project Outcomes

In the immediate term, the project successfully delivered a coherent physical product concept supported by parametric CAD models, fabrication-aware design decisions, and a polished animated product showcase. Despite the absence of a fully functional electronic build, the outputs effectively communicated form, interaction intent, and brand identity.

In the longer term, the most significant outcomes were the acquisition of deep, practical knowledge in areas such as parametric modelling discipline, consumer-grade 3D printing behaviour, tolerance management, and interdisciplinary workflow coordination. The project also fundamentally shifted the author’s approach to learning, highlighting the value of early physical experimentation over extended theoretical research when working with fabrication-centric tools.

Future Development and Project Extension

If the project were to be extended, the next phase would prioritise completing a fully functional physical build, integrating electronics and validating long-term assembly robustness. A custom PCB would likely be developed for a premium variant to consolidate internal complexity and reduce assembly friction. Further refinement of fastening strategies, enclosure tolerances, and material finishes would also be undertaken.

From a process perspective, future iterations would adopt a more strongly staged development structure, introducing earlier physical prototyping to prevent prolonged abstraction. This would reduce the likelihood of over-researching and accelerate convergence toward viable solutions.

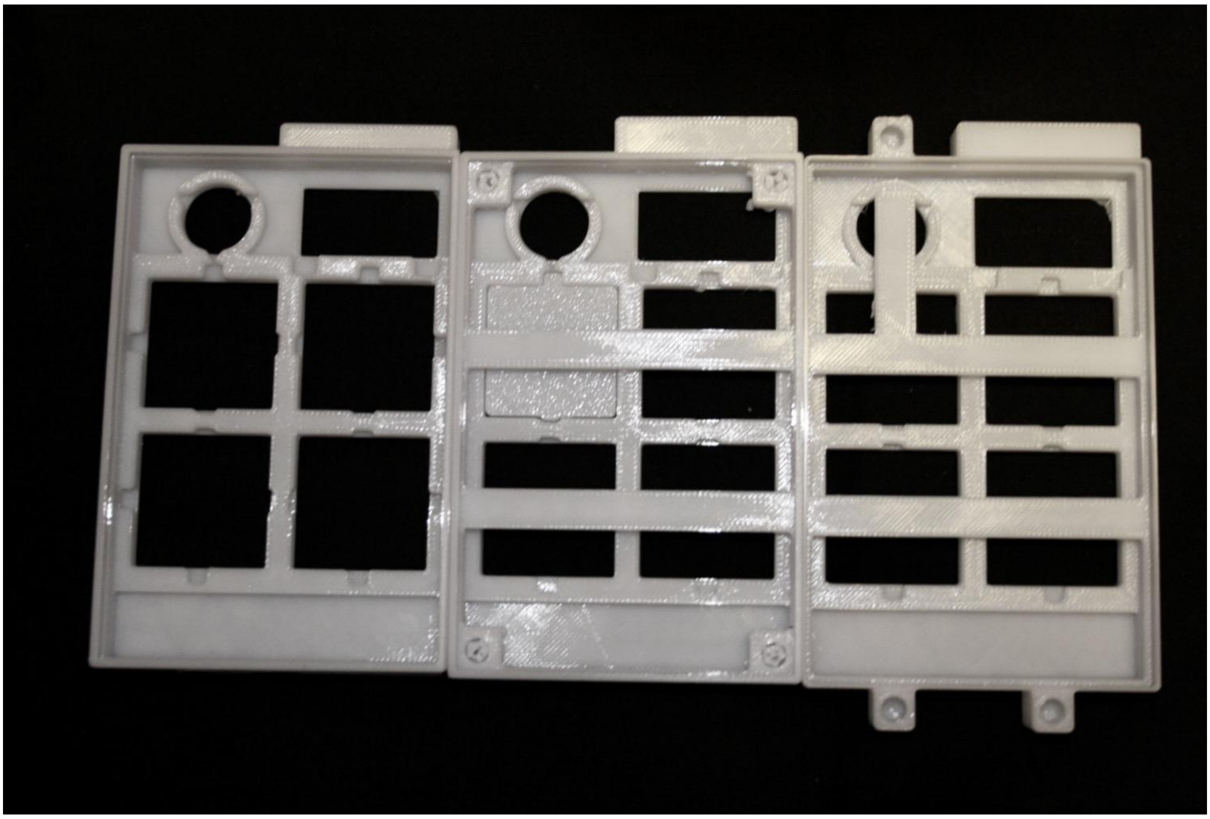
Short-Term Building and Brand Development

In the short term, additional physical builds would enable validation of ergonomics, durability, and repeatability across multiple prints. These builds would also support informal user testing, informing refinements to interaction and assembly flow.

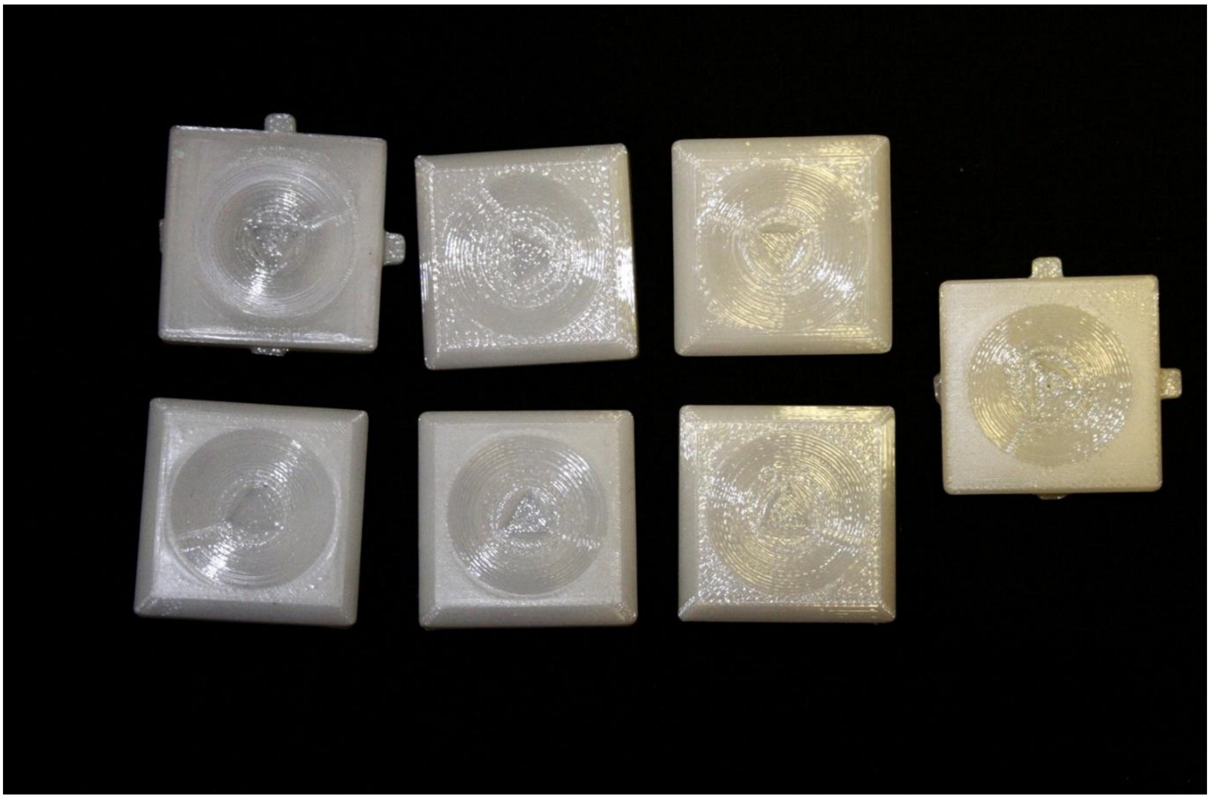
Beyond the single artefact, the brand could be fleshed out through the development of a small family of related products, applying the same visual language and manufacturing-aware principles. This would demonstrate scalability while maintaining the design-led, small-batch ethos established during the project and reinforce the conceptual coherence of the brand ecosystem.

Testing

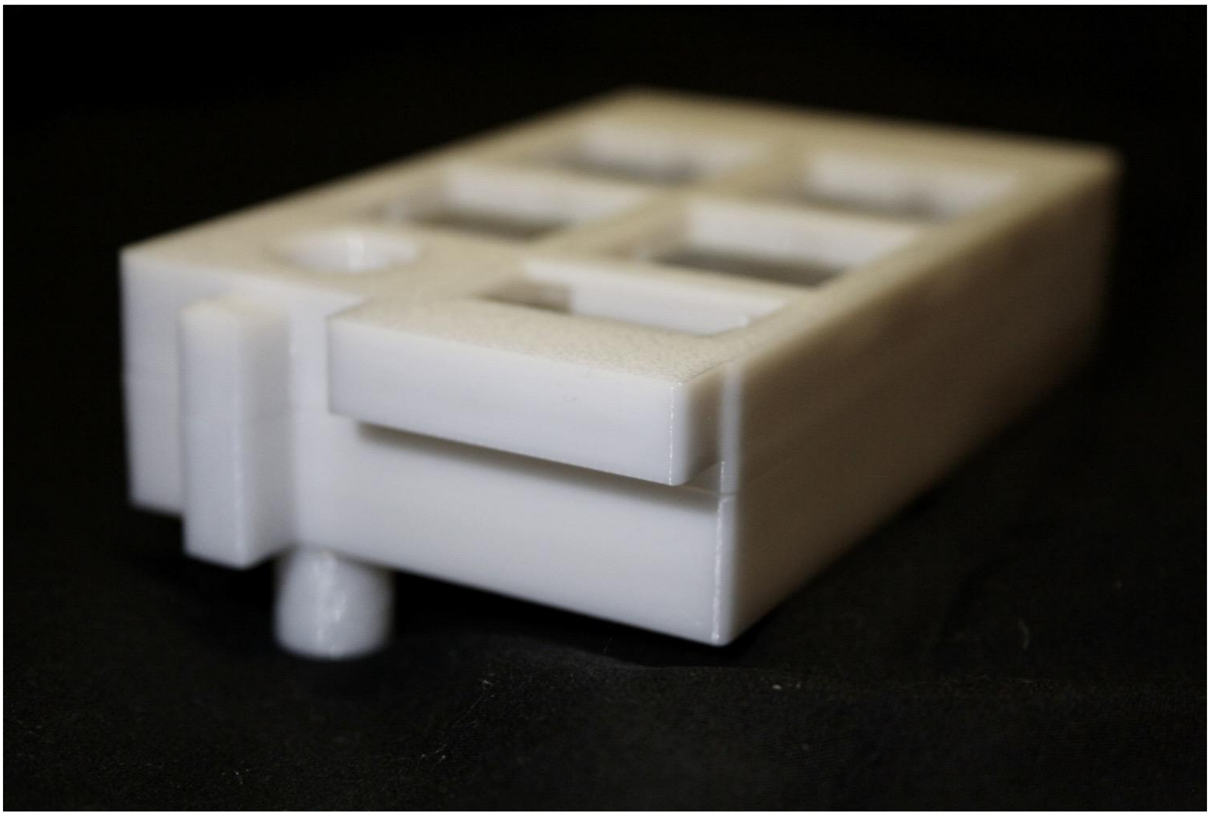
Testing within the project focused primarily on validating design decisions related to parametric CAD geometry, consumer-grade FDM 3D printing, and material behaviour. Rather than formal software or electronics testing, the emphasis was placed on fabrication-aware evaluation, where physical outcomes were used to assess the viability, durability, and usability of the design.



2 Enclosure Testing Iterations



3 Button iterations, testing clearances, layer heights, surface finish, bezels, and fillets



4 Fitment testing

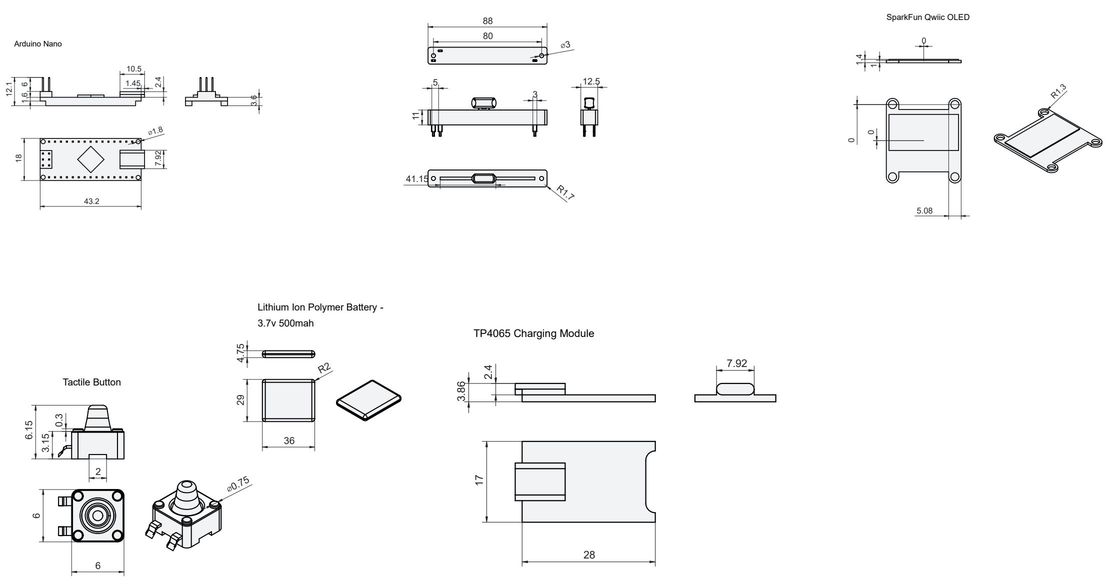
CAD and Clearance Testing (White-Box Testing)

White-box testing was conducted at the CAD level to validate dimensional accuracy, feature relationships, and assembly clearances. Parametric models were iteratively adjusted to test tolerances between interacting elements such as enclosure halves, internal mounting features, screw bosses, and control interfaces. Section views, interference checks, and incremental dimensional changes were used to identify and resolve conflicts before committing to physical prints.

Clearance values were refined through repeated digital iteration and targeted physical validation, particularly in areas where FDM variability was known to affect fit. This process reduced the frequency of failed assemblies and ensured that design intent translated reliably into fabricated parts.

Technical drawings

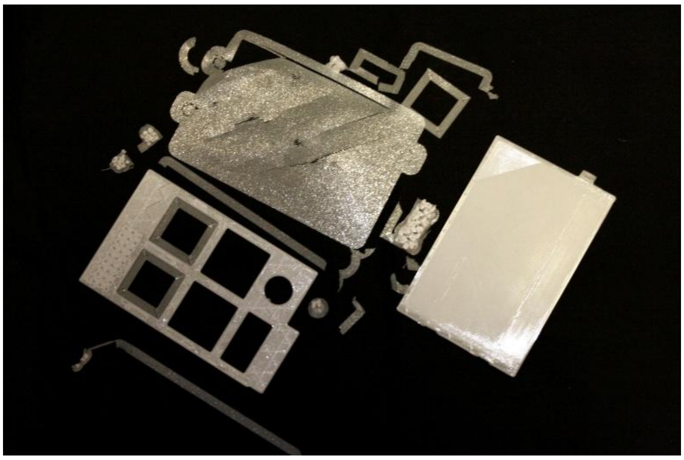
To aid testing and iterating on the design, I created multiple technical drawings in FreeCAD of the internal components. This allowed for quick reference during enclosure design and was on of the first steps taken during CAD development. While they proved useful, they were also intensive to render on my machine and crashed my software multiple times during modelling.



Print Calibration and Adhesion Testing

Extensive testing was carried out to optimise print reliability and consistency. Variables such as bed temperature, nozzle temperature, first-layer height, print speed, cooling behaviour, and surface preparation were adjusted iteratively to improve adhesion and reduce print failures. Adhesion issues were treated as a recurring test condition rather than isolated failures, with solutions validated across multiple prints.

Calibration prints were used to refine extrusion accuracy and layer consistency, reducing dimensional drift and improving repeatability. These tests directly influenced design decisions related to wall thickness, feature sizing, and tolerance allowances.



5Print failures from improper bed adhesion

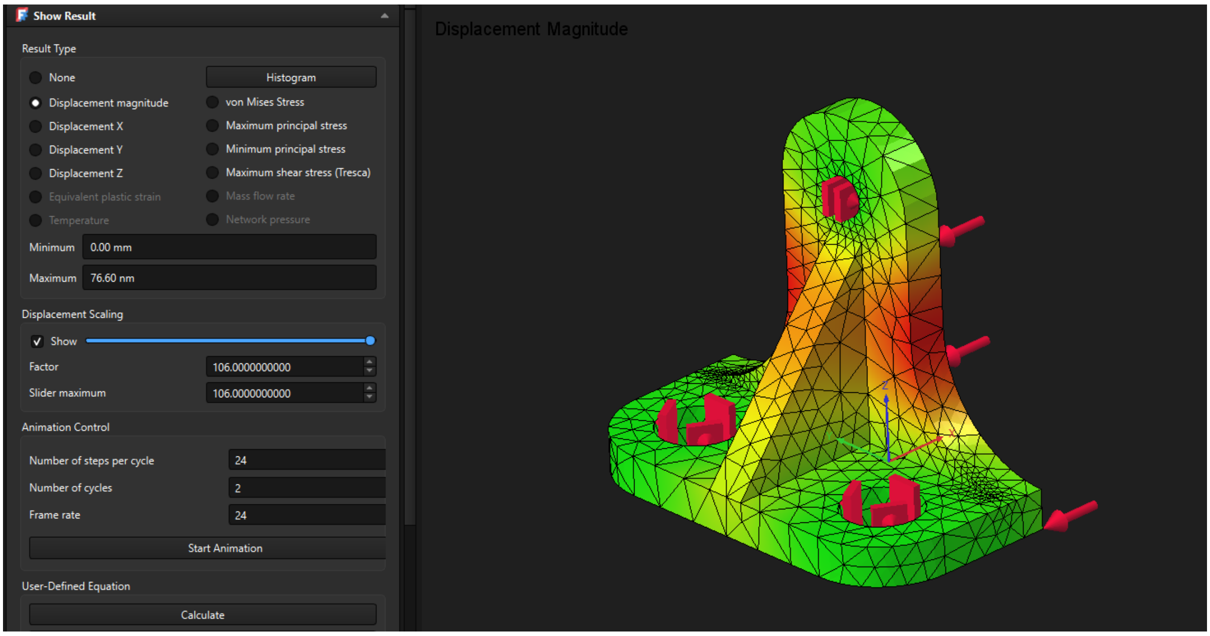
6 Print Warped from improper bed adhesion, incorrection nozzle and bed temperatures, and extended periods of cooling

7 Brims and skirt loops proved effective for reducing bed and warp related issues

Structural Testing and FEM Analysis

Basic structural testing was conducted through a combination of simulated and physical methods. Finite Element Method (FEM) analysis was used at a high level to visualise stress distribution around fastening features, thin walls, and high-load areas such as screw bosses. While not intended as an engineering-grade validation, this analysis supported informed decisions regarding wall thickness, filleting, and reinforcement placement.

Physical stress testing complemented this analysis through drop tests and handling trials. Printed parts were subjected to repeated handling and controlled drops to assess crack initiation, layer separation, and failure points. Infill patterns and densities were varied to evaluate their effect on strength-to-material ratios, informing decisions around internal structure and print efficiency.



8 Example fem test carried out on learning model

Black-Box Testing and Usability Evaluation

Black-box testing focused on perceived usability, assembly logic, and interaction clarity rather than internal implementation. Assembly flow was evaluated through repeated test fits, assessing ease of alignment, fastening sequence, and tolerance robustness. Visual clarity and interaction affordances were evaluated through the animated product showcase, which functioned as a proxy for user testing by revealing potential ambiguities in form, orientation, and control placement.

This combination of physical testing and visual evaluation ensured that the design remained legible, functional, and plausible despite the absence of a fully operational electronic prototype.

Conclusion

This project demonstrated how a coherent, design-led hardware concept can be developed within real manufacturing constraints through the integration of parametric CAD, fabrication-aware design, and digital presentation. By prioritising physical plausibility, abstraction, and brand coherence over full electronic implementation, the project delivers a credible physical prototype supported by animation and web presentation.

The work highlighted the importance of designing with manufacturing processes in mind, particularly within the limitations of consumer-grade FDM 3D printing. Iterative modelling, material testing, and physical prototyping grounded design decisions in technical reality and facilitated meaningful skill development across previously unfamiliar tools.

While scope was deliberately reduced, this refinement enabled greater depth, learning, and quality of execution. Overall, the project met its objectives and provided a strong foundation for future exploration in design-led hardware development and small-scale digital fabrication, and I imagine in the future I will be using the skills developed to great effect.

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Major Learning Resources for FreeCAD and Blender:

https://youtu.be/BS7MK5nqNrw?si=gxsHabK9xmLxzeNb\

https://youtu.be/5cOHqoIqQnw?si=NjlpVFJ5Jn_ZPLMC

https://youtu.be/LTOmmwzBEa8?si=QbO2jtqRxPHkF3rl

https://youtu.be/QO58XGnc9Xk?si=6v4IPA4hQv30QWab

https://youtu.be/43t_yq8AuXY?si=vY0XbVlXC6fAlyOA

https://polyhaven.com/a/brown_photostudio_02

https://blender.stackexchange.com/questions/293661/loading-render-kernels-what-exactly-that-means

https://blender.stackexchange.com/questions/5519/setting-the-pivot-point-of-an-object

https://artisticrender.com/?s=material

https://polyhaven.com/a/green_metal_rust

https://docs.blender.org/manual/en/latest/files/linked_libraries/link_append.html

https://shunpoly.com/article/how-to-create-plastic-in-blender

https://whatmakeart.com/3d-modeling/blender/install-blenderkit-blender/

https://github.com/BlenderKit/blenderkit/wiki/BlenderKit-add-on-documentation#assetbar

https://polyvia3d.com/repair/obj

https://www.reddit.com/r/blender/comments/s0vkfd/i_cant_bevel_my_edges_is_there_something_im_doing/

https://youtu.be/S2ySQaKKiHA?si=7gpfHGe-5e-X1EgM

https://youtu.be/ZPsLhvgU8kc?si=1YtE7PwI67YzoSWz

https://youtu.be/p2yDGlv5wl0?si=KrvCXQIFW_i_Kmup

https://youtu.be/YJd78HWdK1M?si=4choAvELAJCXlj7j

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https://youtu.be/MvH5NsDccTc?si=NmZ4FPWEzWi1Zbcj

https://youtu.be/3vGaiArjKko?si=_YL2MTsuFBoaFkbf

https://youtu.be/z2o4hmli_10?si=1NTCSsJfZNQMP_Ve

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https://youtu.be/XKrDUnZCmQQ?si=VDydZSslcSJgDcu8

https://youtu.be/b2JFQLStJTc?si=ENd_vo5U2SjQKZF4

https://youtu.be/v689Ii71TnE?si=Y2zfhgPn8C9Hp8xy

https://youtu.be/zOsAy-8j0wg?si=YvKrwxeIoPdbj-Mg

https://youtu.be/v6FgTIpsCKo?si=TIEId7k-1x9PL9ls

https://youtu.be/krrqydtneO0?si=HIrdmrkcT1bj6H4_

https://youtu.be/EFD556ysIUg?si=sGuejBe02yj6oSg7

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